Karim Khalil

Computer Vision / AI Engineer

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PROFESSIONAL SUMMARY

AI engineer focused on computer vision, with 2+ years of experience developing, training and deploying YOLO-based object detection models for live video and aerial imagery. Experience serving models through FastAPI, containerizing them with Docker, and building the web front ends that present the results. Projects include PPE compliance, restricted-area intrusion, people counting, driver fatigue, fire and smoke detection, and palm counting from aerial imagery.

TECHNICAL SKILLS

Computer Vision: YOLOv8, YOLO26, object detection, OpenCV, real-time video processing, CNNs, activity recognition
Model Development: Training and validating models on custom datasets (LabelImg), transfer learning, inference optimization
ML Frameworks: Python, PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Dlib
Deployment and APIs: FastAPI, Flask, REST APIs, Docker, multi-threading, Postman, Git, GitHub
Web and Geospatial: HTML, JavaScript, Leaflet maps, KML farm boundaries, satellite and aerial imagery
AI Tooling: Claude Code, LLM applications, OpenClaw, Replit, SQL

PROFESSIONAL EXPERIENCE

Platform Operation & Data Analyst | Platfarm | Royal Commission for AlUla (RCU)
Jun 2026 - Present | AlUla, Saudi Arabia (On-site)

- Built Date Palm AI: trained a custom YOLO26 model that counts date palms in aerial imagery per farm polygon (7,124 farms, 16 regions), with a FastAPI backend, upload page and Leaflet farm map.
- Managed platform operations and validated dashboards and reports to ensure data accuracy, quality and consistency.
- Built AI applications and automated workflows using Python, Replit, OpenClaw and LLM technologies, including a WhatsApp reporting assistant.

AI Engineer | Nasam Technology
May 2025 - Apr 2026 | Raidah, Saudi Arabia (Remote)

- Built and deployed AI-powered APIs using FastAPI so internal systems can consume real-time inference results.
- Optimized real-time video processing pipelines for environmental monitoring, improving response latency by about 15%.
- Tested and validated AI services using Postman to ensure consistent performance and reliability across deployments.

AI Engineer | Omni Clouds
Jul 2024 - Feb 2025 | Dubai, UAE (Remote)

- Developed and deployed computer vision models for real-time video monitoring, containerized with Docker for internal production use.
- Implemented YOLO-based object detection pipelines for live video streams, improving detection reliability and reducing false alerts by about 20%.
- Collaborated with software engineers to integrate AI models into internal systems and applications, supporting continuous real-time inference and system stability.

PROJECTS

Date Palm Detection & Counting (YOLO26)

- Trained a custom YOLO26 model to detect and count date palms in aerial and satellite imagery, with an upload page and a farm map showing counts per farm.
- Measures palm size per image and tiles inference to keep counts consistent; counts only palms inside each farm boundary.
- Tools: Python, YOLO26, PyTorch, OpenCV, FastAPI, Leaflet, KML

PPE Detection System

- Implemented YOLOv8 object detection models to monitor PPE compliance in industrial environments.
- Trained and validated models on custom datasets, reducing manual monitoring by about 30%.
- Tools: Python, PyTorch, YOLOv8, OpenCV, LabelImg

Restricted Area Intrusion Detection System

- Developed an AI security system using object detection and motion analysis for factory surveillance.
- Generated real-time alerts with visual overlays, improving response time by about 25%.
- Tools: Python, OpenCV, YOLOv8, NumPy, Flask

People Detection and Counting in Restricted Zones

- Detected and counted people inside a restricted zone from video and saved the captured images to a folder.
- Tools: Python, OpenCV, object detection

Driver Inattention Monitoring System

- Developed a real-time driver monitoring system using CNNs and facial landmark tracking to detect fatigue and distraction.
- Optimized inference for continuous real-time processing, achieving about 20% faster detection responsiveness.
- Tools: Python, PyTorch, OpenCV, Dlib, multi-threading, NumPy

Human Activity Detection System

- Built activity recognition models using transfer learning on temporal video data for elderly safety and posture monitoring, with real-time inference suitable for continuous alerting.
- Tools: Python, TensorFlow, Keras, OpenCV, Scikit-learn

Fire and Smoke Detection System

- Engineered a CNN-based classifier for early fire and smoke detection across varied environments, optimized for low-latency deployment in surveillance pipelines.
- Tools: Python, TensorFlow, Keras, OpenCV, Pandas, NumPy

EDUCATION

B.Sc. in Computer Science | Canadian International College (CIC) | 2024

CERTIFICATIONS

AI For Everyone - Coursera; Data Analysis - egFWD; GIS Professional - NARSS; Competitive Programming Level 1 - Coach Academy EG; Communication Skills - Canadian International College

LANGUAGES

Arabic (native), English (professional)